IFTW Logger

The assistant in USBC PC host applications

About this document

Scope and purpose

This automation tool assists users in collecting all necessary information / data through Infineon CY4500 CC sniffer and 3rd party log devices, includes Saleae and Ellisys C-Tracker.

Intended audience

Users who adapt Infineon USB Type-C solutions in PC HOST applications and frequently communicate with Infineon SYS support teams in issue debugging discussions.

Table of Contents

1 Introduction

Infineon USBC CCG devices are commonly used in PD port controller applications in systems like Notebook and Desktop computers, tablets, mobiles etc. In such cases, the system design typically includes a Host Processor or Embedded Controller (EC) that is responsible for the system policy decisions. The CCG controller can be connected to the EC and the CCG firmware allows the EC to monitor the status of the USB-PD ports, change configurations, perform firmware updates, and transparently interact with other USB-PD devices connected to CCG. All the capabilities are exercised through commands/responses transferred across the Host Processor Interface (HPI).

When seeing issues in the field, it is required to provide relevant information to Infineon SYS support team to accommodate preliminary diagnosis and appropriate resolutions. Therefore, a comprehensive log package with accurate data is mandatory to accelerate solutions delivery.

Figure 1 shows the standard procedure of debugging flows:

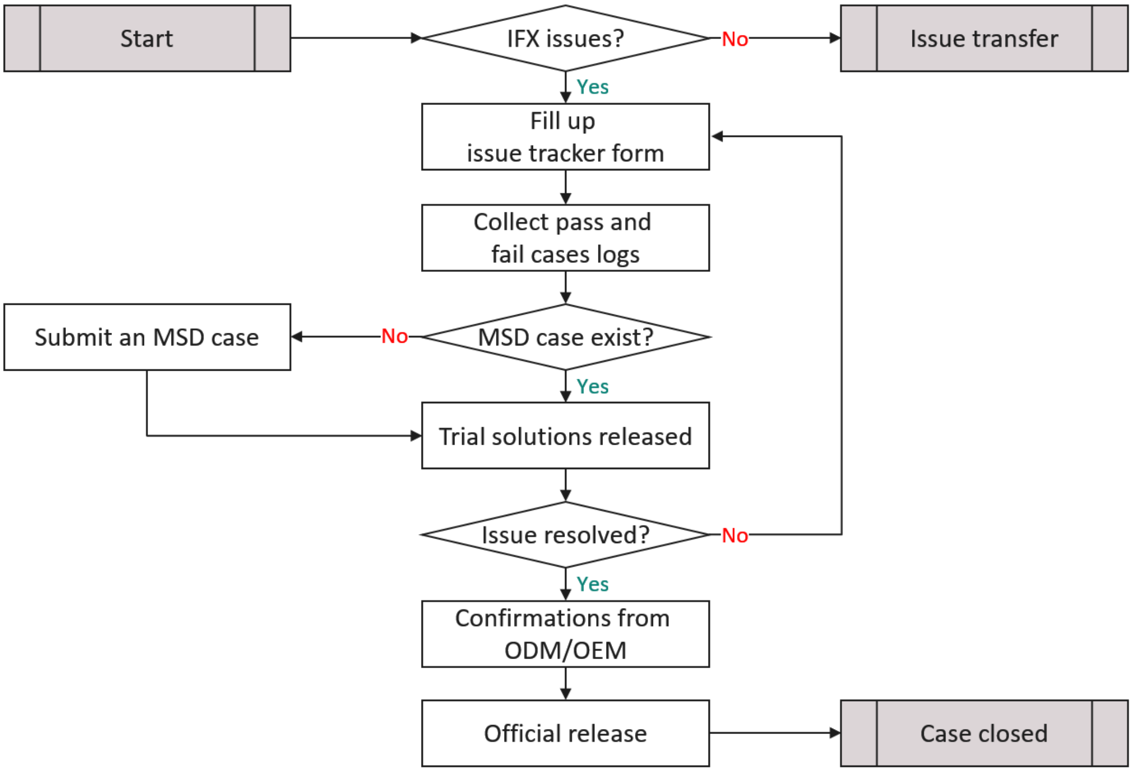


Figure 1 Standard issue report flow

1.1 Prerequisites

There are plenty of log devices available in the market. Except Infineon self-own CY4500 CC sniffer, there are two 3rd party tools recommended to enhance debugging efficiency:

1. Saleae Logic Pro 16 + Saleae Logic2 application v2.4.0 or above
2. Ellisys C-Tracker + Ellisys Type-C Tracker Analyzer

Infineon SYS support team creates a tool package to integrate all supported log devices into a single application, alone with advanced plugins to facilitate log collection activities. It includes:

- **HPI Analyser plugin for Saleae**: a specialised I²C analyser dedicated to Infineon HPI protocol.

- **CC Analyser plugin Saleae**: USB Type-C vendor defines message analyser.
- **Infineon IFTW Logger**: a GUI tool to generate logs with all necessary information.

1.2 Installation

1. Install EZ-PD Protocol Analyzer Utility v4.2.0 (download here).
2. Install Saleae Logic2 application (www.saleae.com) and:
   a. Enable "***Automation***" via Preference -> Automation -> Enable Automation Server.

Figure 2 Saleae Logic2 Preferences – Automation

   b. Copy "***HPI_regs.xml***", "***i2c_analyzer_for_HPI.dll***" and "***saleae_pd_analyser_release.dll***" to the folder assigned in "Custom Low Level Analysers".

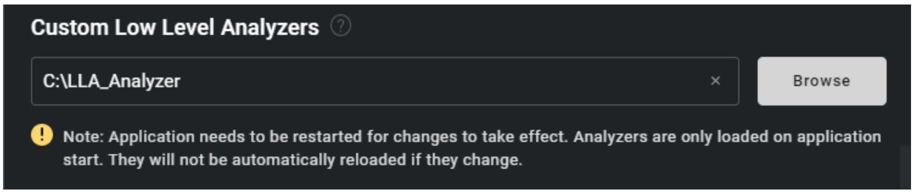
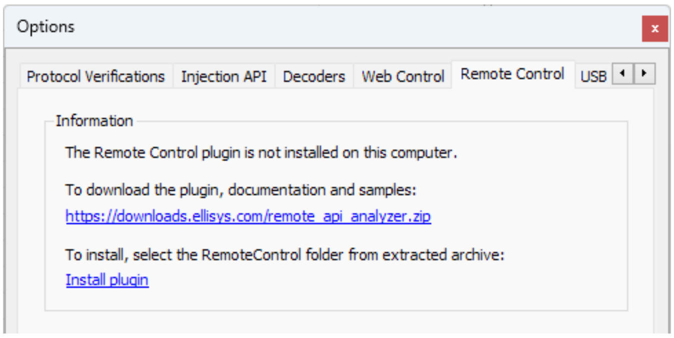
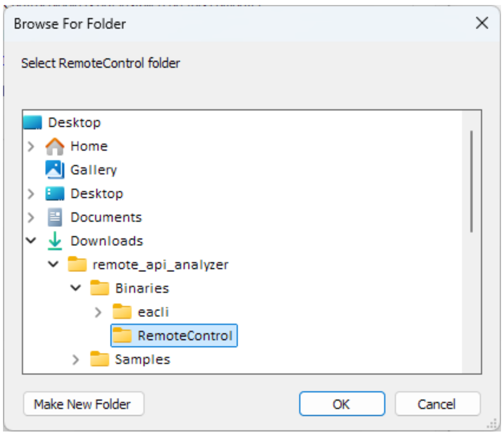


Figure 3 Saleae Logic2 Preferences – Low Level Analysers

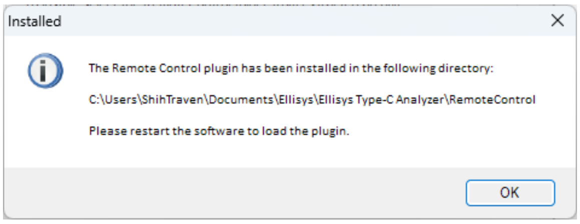
3. Install Ellisys Type-C Tracker Analyzer application (www.ellisys.com) and:
   a. Download plugin from prompted link.



   b. Unzip and install via the "Install plugin".



   c. Restart Ellisys C-Tracker application.



d. Enable Remote Control under Tools > Options > Remote Control. When running this tool on the local machine, the TCP/UDP port setting is irrelevant and can be set with random number.

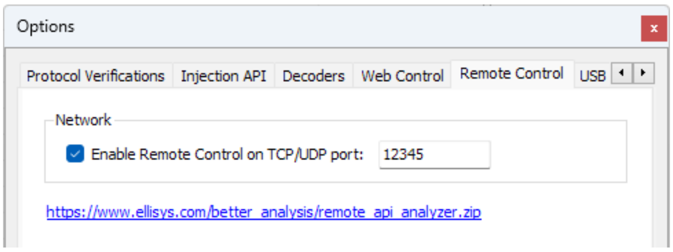


Figure 4 Ellisys Type-C Tracker Analyzer Remote Control

4. Visit GitHub (https://github.com/IFTWTraven/iftw_logger) repository and download "**binary**" folder to arbitrary directory except root directory, which might need additional access permission for file creation later. Current user's desktop is recommended.

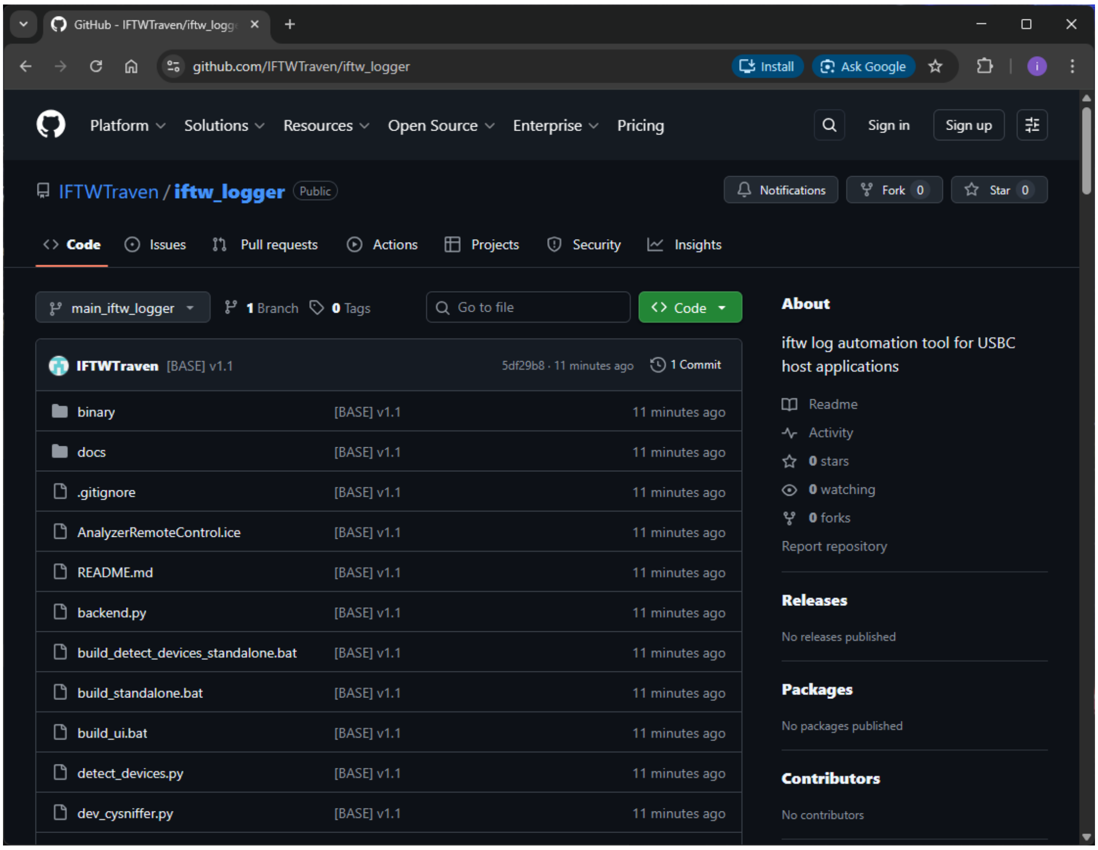


Figure 5 IFTW logger tool package

5. Done.

2 Infineon Automation Tool Package

2.1 HPI Analyser

The native I²C Analysers in Saleae Logic2 application can decode and list bus messages to its terminal-like window and tabular-format table for search usage. It decodes I²C commands in BYTE format basis and is useful in generic use cases.

However, Infineon HPI protocol is a dual-byte basis as its register address and usually conducts a WORD or longer data length as a single command and responses, and it is not intuitive to translate decoded messages through native I²C analyser. Worse, it is hard to do HPI command search. Given that it causes additional efforts and inconveniences to users, especially for engineers who are dealing with HPI activities in daily basis.

Hence, Infineon SYS support team creates a customised Saleae I²C analyser plugin. It can re-synthesise standard I²C byte packets to scalable-length message format, allows users to change significant byte order and capable to translate HPI raw data. In addition, it is easier to do HPI command analysis by introducing timestamp ahead of every HPI messages.

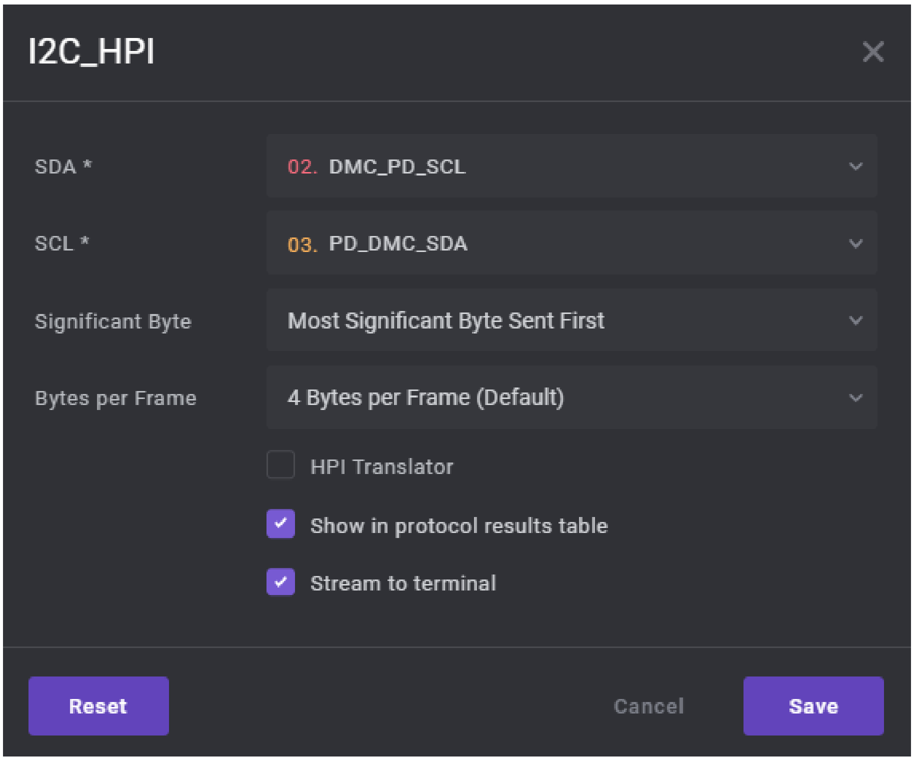


Figure 6 Additional options in Infineon "I2C_HPI" analyser

2.2 CC Analyser

USB CC and HPI logs are mandatory information Infineon SYS support team usually need while dealing with issues. In the past, they were provided by CY4500 CC sniffer and Saleae Logic Analyser separately and always was difficult to understand command sequences among CC and HPI activities because these logs are not referring the same timeline.

By introducing Infineon CC Analyser with a proper configuration, users can capture and decode both USB CC and HPI logs through Saleae Logic2 application alone. It is extremely useful to check relevant command sequences across different protocols and accelerate debugging progress.

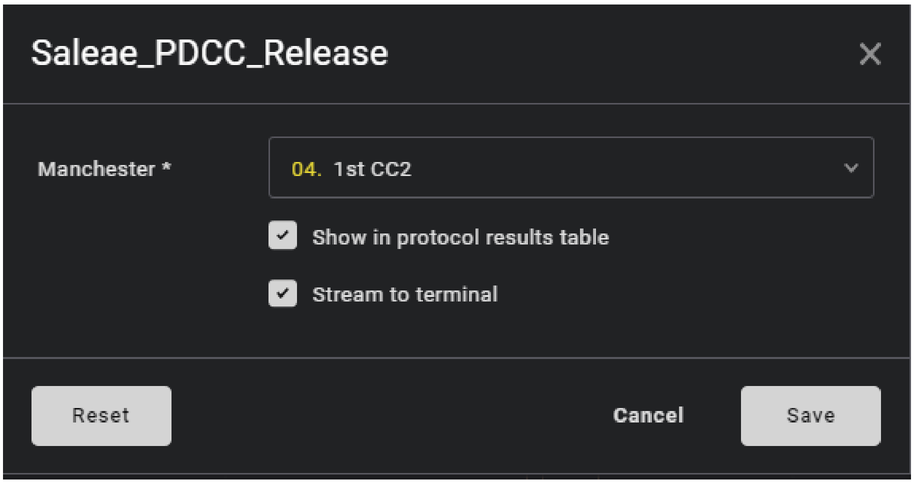


Figure 7 Infineon "Saleae_PDCC_Release" Analyser

2.3 Infineon IFTW Logger

This is a portable application that supports Microsoft Windows 10 and above. It is highly recommended to use it with Saleae Logic Analyser Pro 16 and Ellisys C-Tracker. The application pre-defines all necessary trace pins that users are required to provide while reporting issues for both AMD and INTEL design cases, along with project-relevant information, such as PD firmware revision. All numbers in pulldown menus are selectable but cannot be duplicated.

Category	Host Platform		CY4500	SALEAE			Ellisys
	AMD	INTEL	Pins	Channel	Mode	Analyser	C-Tracker
Generic	CC1	CC1	O	0	analog + digital	Manchester_PD_CC	O
	CC2	CC2	O	1	analog + digital	Manchester_PD_CC	O
	SBU1	SBU1	O	2	analog + digital	-	AUX
	SBU2	SBU2	O	3	analog + digital	-	AUX
	VBUS	VBUS	O	4	analog + digital	-	O
	PD UART	PD UART	-	5	digital	Async	UART
	EC-PD SDA	EC-PD SDA	-	6	digital	HPI_I2C	I2C
	EC-PD SLK	EC-PD SLK	-	7	digital	HPI_I2C	I2C
	EC-PD INT	EC-PD INT	-	8	digital	-	I2C
SoC	APU-PD INT	PCH-PD INT	-	9	digital	-	2nd I2C
	APU-PD RST	PCH-PD SDA	-	10	digital	HPI_I2C (INTEL)	-
	APU-PD SDA	PCH-PD SLK	-	11	digital	HPI_I2C	2nd I2C
	APU-PD SLK	BBR PWR	-	12	digital	HPI_I2C (AMD)	2nd I2C
Mux	MUX PWR / HPD	BBR RST	-	13	digital	-	-
	MUX-PD SDA	BBR-PD SDA	-	14	digital	HPI_I2C	2nd I2C
	MUX-PD SLK	BBR-PD SLK	-	15	digital	HPI_I2C	2nd I2C

Figure 8 Pin needs per USBC PD Host E2 training

SALEAE/Ellisys log files will be generated based on channel number settings, associated with corresponding bus analysers respectively. These files are named beginning with DATE_TIME as a prefix and appended with selected data in the Issue Information section. In addition, an issue tracker form and a text file named by the current date will be outputted under the same naming folder.

The Main window splits in following sections (refer to Figure 9):

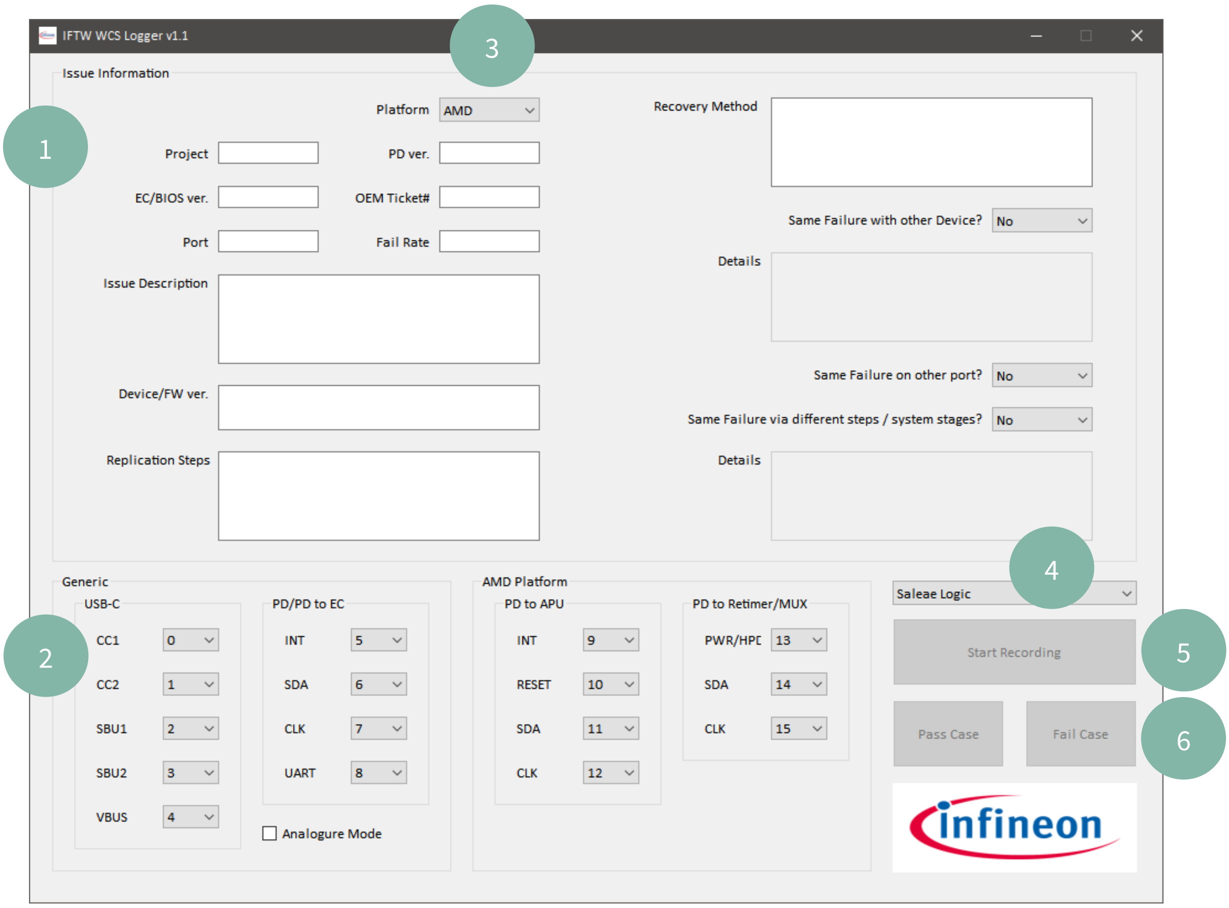


Figure 9 Infineon IFTW Logger

1. **Issue Information**: it is mandatory to fill all information correspondingly, whereas OEM Ticket# as optional. It also covers both AMD and INTEL design trace needs. All information will be recorded into issue tracker form automatically.
2. **Generic**: this section describes common trace pins need in USBC PC host application. This section will be switched to Ellisys mode while work in “Ellisys C-Tracker” mode (Figure 10):

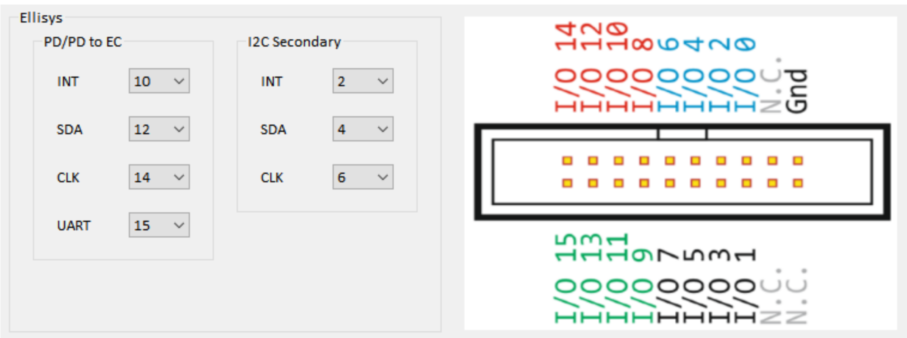


Figure 10 Ellisys mode

   a. **USB-C**: it defines SALEAE channel numbers to basic USB Type-C pins. All channels are recorded in digital mode as default, and can be set to digital + analogue mode by enabling the “Analogue Mode” option if necessary.

   b. **PD/PD to EC**: it defines HPI bus to EC and PD UART.

3. **AMD/INTEL Platform**: it defines all necessary trace pins need for AMD and INTEL designs respectively. Please be notices that due to insufficient pin counts available in SALEAE, in AMD designs, MUX power pin SALEAE channel is shared with HPD pin. Please connect either one depends on issue category for AMD design cases.
4. **Saleae Logic/Ellisys C-Tracker**: it supports both log devices and will refresh its selections while application is launched. In addition, EZ-PD Protocol Analyzer Utility will incorporate with Saleae Logic2 application when work in “Saleae Logic” mode.
5. **Start/Abort Recording**: the main button to control log applications. It will be grey-out if data missing in Issue Information section or channel selection duplication. Use this button to start / stop log capturing processes.

6. **Pass/Fail Case**: the main buttons to save recorded logs for pass and fail cases. By clicking either buttons, it will save the current logs as pass or fail cases and export to an issue tracker form with all data described in Issue Information section. It will be grey-out before log capturing.

3 How to use

Figure 11 is functional flow chart to illustrate Infineon IFTW logger working model. IFTW Logger will run into either Saleae or Ellisys mode based on connecting status of both SALEAE Logic Analyzer and Ellisys C-Tracker Analyzer while launching. Please notice that due to log file requirement from Infineon SYS support teams, it is mandatory to connect CY4500 if work in Saleae mode.

Depends on log applications running condition, there are 2 possible scenarios in Infineon IFTW Logger usage:

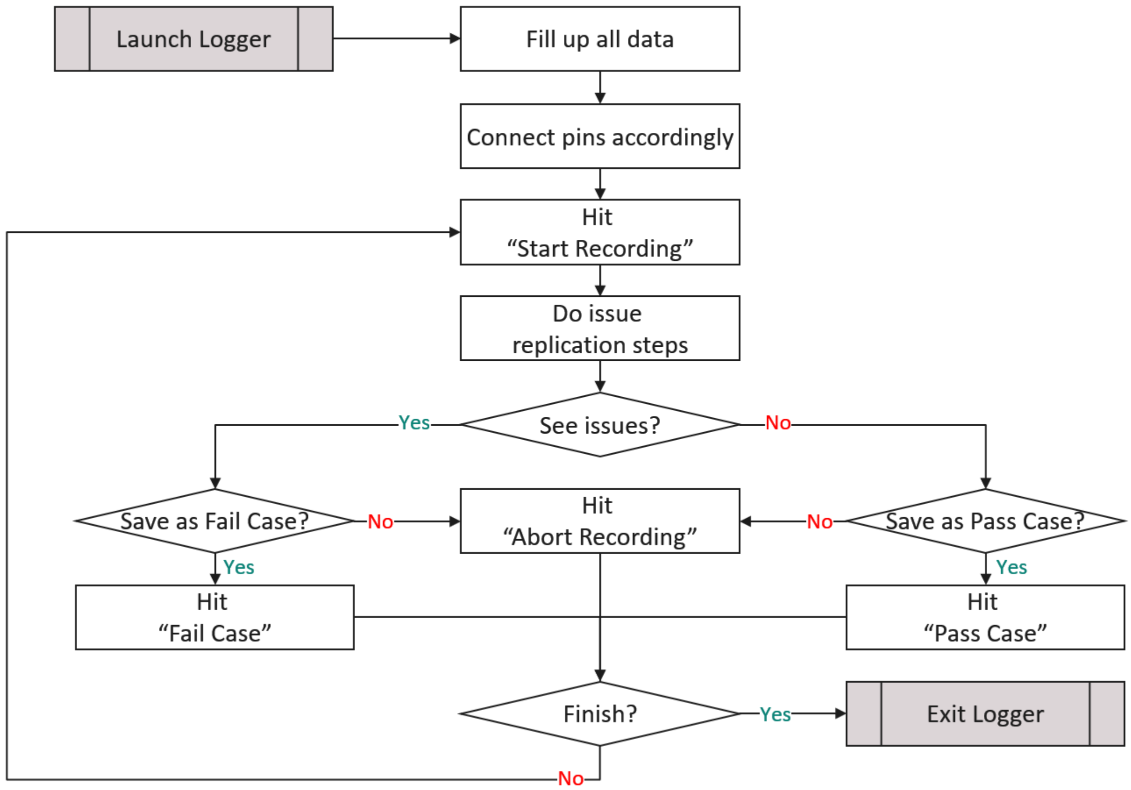


Figure 11 Infineon IFTW Logger functional flow

3.1 Scenario A

This user scenario presumes either Saleae Logic2 or Ellisys C-Tracker application is not running in the background while starting Infineon IFTW Logger application. The benefit is users do not need to enable Saleae Automation option beforehand.

1. Start Infineon IFTW Logger.
2. Fill all information and platform type in Issue Information and mapping pins with pre-defined channels or preferred ones.
3. Click "Start Recording" to start log capturing per setup.
4. Infineon IFTW Logger launches log applications with automation supportive.
5. Do issue replication steps. (e.g., shutdown and reboot to S0, re-plugging devices again, etc.)
6. Click "Abort Recording", "Pass Case" or "Fail Case" while replication steps finished.
7. Infineon IFTW Logger will

    a. In Saleae mode: assigns bus analysers to the SALEAE log and save. Record pin capture settings in the text configuration file. In additional, a CY4500 log is created and save.
    b. In Ellisys mode: save log file and close it.
    c. Update tracker form sheet with all recorded log filenames.
8. Repeat step 3 to 7 for another log capture round.
9. Close Infineon IFTW Logger while finished, and log applications will be shut down.

3.2 Scenario B

This user scenario only works while automation option is enabled in Saleae Logic2 application as pre-requisites in Saleae mode. In this case, it is presumed users launch log applications already.

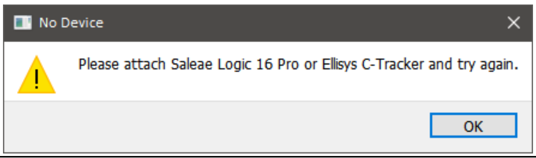
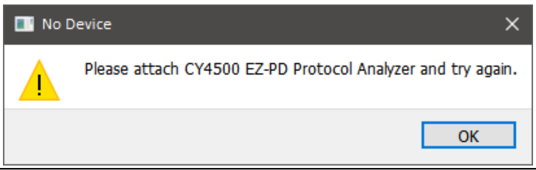
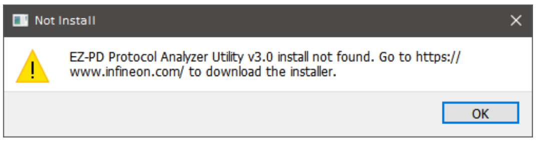
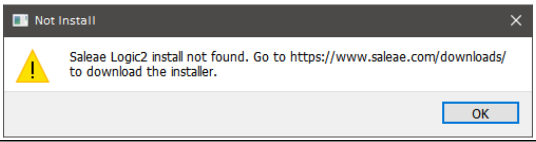
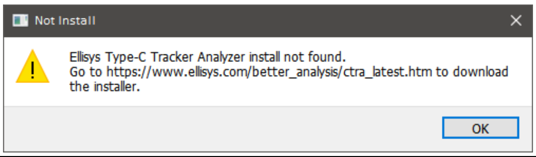
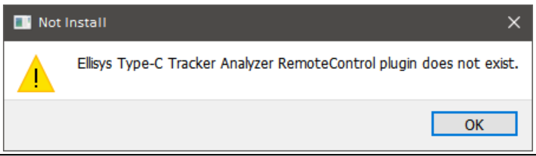
1. Start Infineon IFTW Logger.
2. Fill all information and platform type in Issue Information and mapping pins with pre-defined channels or preferred ones.
3. Click “Start Recording” to start log capturing per setup.
4. Do issue replication steps. (e.g., shutdown and reboot to S0, re-plugging devices again, etc.)
5. Click “Abort Recording”, “Pass Case” or “Fail Case” while replication steps finished.
6. Infineon IFTW Logger will
    a. In Saleae mode: assigns bus analysers to the SALEAE log and save. Record pin capture settings in the text configuration file. In additional, a CY4500 log is created and save.
    b. In Ellisys mode: save log file and close it.
    c. Update tracker form sheet with all recorded log filenames.
7. Repeat step 3 to 6 for another log capture round.
8. Close Infineon IFTW Logger while finished, and log applications will ***NOT*** be shut down.

4 Limitations

There are several limitations of Infineon IFTW Logger, including:

1. It supports Microsoft Windows 10 and above only.
2. It verified on English and Traditional Chinese OS only.
3. It might crash if Saleae Logic2 application does not enable automation option while working at Scenario B.
4. This tool relies on Windows UI automation to support Saleae channel renaming and control the CC Sniffer application. Direct user interaction with these applications can disrupt the automation; in particular, minimizing their windows may cause the Infineon IFTW Logger to crash. To ensure stability, avoid interacting with these applications while the tool is running.
5. Known file saving issue with Ellisys C-Tracker Analyser version 1.0.9503.
6. It does not support Saleae Logic Analyser 8 / Pro 8 due to insufficient channel numbers.

Self Service

CATEGORY	SYMPTOM	TROUBLE SHOOTING
Not Start.	IFTW Logger does not start and shows one of following dialogs instead.  	Please check if either Saleae Logic 16 Pro or Ellisys C-Tracker is being appropriate attached and enumerated on Host systems or not.
	 	Please check if CY4500 is setup as PD3.0 supportive and being attached and enumerated appropriately on Host systems or not.
	 	Please go to Infineon website and download latest EZ-PD Protocol Analyzer Utility installer. It is highly recommended to install v4.0 or above by its default setups, to support both SPR and EPR sniffers.
	 	Please go to Saleae website and download latest Logic2 installer. It is highly recommended to install this application by its default setups.
	 	Please go to Ellisys website and download latest Ellisys Type-C Tracker Analyzer installer. It is highly recommended to install this application by its default setups.
	 	Follow **1.2 Installation** section and setup remote control plugin.
IFTW Logger crashes.	IFTW Logger crashes after hitting “Start Capturing”	Please check if Saleae Logic2 application is running and did not enable its automation option while launching IFTW Logger.
		Please check if EZ-PD Protocol Analyzer Utility is stuck at file saving or save confirmation dialog. It is recommended to close all popup dialogs or shutdown EZ-PD Protocol Analyzer Utility before launching IFTW Logger.
		Please check if latest version of EZ-PD Protocol Analyzer Utility is installed. It is recommended to install v3.1.x with default setups.
		Please check if “tracker_form.xlsx” is placed in the same folder with IFTW Logger application or not.
		Please check if “AnalyzerRemoteControl.ice” is placed in the same folder with IFTW Logger application or not.

CATEGORY	SYMPTOM	TROUBLE SHOOTING
	IFTW Logger crashes after hitting either "Pass case" or "Fail Case"	Please check if IFTW Logger is copied to a folder which current user has sufficient permissions to access it.

Revision history

Document version	Date of release	Description of changes
1.0	2023-07-20	1. First release.
1.1	2026-03-13	1. IFTW Logger v1.1 release. 2. Add EPR CY4500 support. 3. Improve SALEAE channel renaming performance.

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